A
Project Report
on

GRAM CONNECT
AI Rural Complaints Management System

Submitted By

A.GEETHA SREE [R210780]
LEKKALA SANJANA [R210570]

Under the Esteemed Guidance of
S.RAJESWARI

Department of Computer Science and Engineering

Rajiv Gandhi University of Knowledge Technologies
(RGUKT) R K Valley, Y.S.R. Kadapa, Andhra Pradesh

I

Rajiv Gandhi University of Knowledge Technologies
(A.P.Government Act 18 of 2008)

RGUKT IIIT RK Valley
Vempalli, Kadapa, Andhra Pradesh-516330

CERTIFICATE OF PROJECT COMPLETION

This is to certify that I have examined the thesis entitled **"Gram Connect -AI Rural Complaints Management System"** submitted by A.Geetha Sree [R210780] & Lekkala Sanjana [R210570] under our guidance and supervision, in partial fulfillment of the requirements for the award of the **Bachelor of Technology in Computer Science and Engineering** during the **academic year 2025–2026** at **RGUKT-RK Valley**.

To the best of my knowledge, the results embodied in this dissertation work have not been submitted to any university or institute for the award of any degree or diploma.

Project Guide
S.Rajeswari

Head of the Department (CSE)
Dr.Ch.Ratna Kumari
Assistant Professor
Head of the department Computer Science and Engineering
RGUKT RK-Valley

External Examiner

II

Rajiv Gandhi University of Knowledge Technologies
(A.P.Government Act 18 of 2008)

RGUKT IIIT RK Valley
Vempalli, Kadapa, Andhra Pradesh516330

DECLARATION

we, **A.Geetha Sree [R210780] & Lekkala Sanjana [R210570]** , hereby declare that the project report entitled **"Gram Connect-AI Rural Complaints Management System"**, carried out under the guidance of **S.Rajeswari**, is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** during the **academic year 2025–2026** at **RGUKT-RK Valley**.

I also declare that this project is a result of our own effort and has not been copied or imitated from any source. Citations from websites are mentioned in the references. To the best of my knowledge, the results embodied in this dissertation work have not been submitted to any university or institute for the award of any degree or diploma.

Date :

Place : RGUKT, RK Valley

A.GEETHA SREE [R210780]
LEKKALA SANJANA [R210570]

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With Sincere Regards,

A.GEETHA SREE [R210780]

LEKKALA SANJANA [R210570]

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Chapter 1

Abstract

Rural governance in India faces a persistent challenge: citizens in remote villages have no accessible, language-friendly channel to report public infrastructure failures to the government. Problems such as damaged roads, water supply disruptions, electricity failures, and sanitation issues remain unresolved for extended periods due to the lack of a structured and transparent complaint management system.

GramConnect is an AI-based rural complaint management system designed to bridge this gap. The platform allows any rural citizen to register a complaint using a simple form, where the Problem Description field supports both voice input and text entry. Voice complaints are transcribed in real time using OpenAI Whisper, enabling non-literate and low-literacy users to describe issues in their own language. Other form fields such as Name, Village, and Category are filled manually by the user.

Once submitted, the complaint is automatically analyzed by an AI classification engine built using TensorFlow and Scikit-learn. The engine assigns the complaint a category (Road Damage, Water Supply, Electricity, Sanitation, or Other) and a priority level (High, Medium, or Low), and routes it instantly to the relevant government department. Government officers access a dedicated Admin Dashboard built in React.js, where they can view all complaints, play voice recordings, and manage resolution workflows.

A key feature of GramConnect is its certificate-based resolution mechanism. After a complaint is resolved on the ground, the officer generates a digital Resolution Certificate (PDF) from the system, which must be physically signed by the complainant as proof of resolution. The signed certificate is then uploaded to the system, and only then is the complaint officially marked as Resolved. This ensures accountability and prevents false closures.

If a complaint is not resolved within the defined SLA (Service Level Agreement) period, the system automatically escalates it to a Super Admin — a higher-level government officer — who monitors department-wise performance and takes corrective action.

The system is built using React.js for the frontend, Python with Flask for the backend REST API, TensorFlow and Scikit-learn for AI classification, OpenAI Whisper for speech-to-text, SQLite3 with SQLAlchemy ORM for database management, and Python ReportLab for certificate generation.

Tech Stack: React.js, Python (Flask), TensorFlow, Scikit-learn, OpenAI Whisper, SQLite3, SQLAlchemy, ReportLab, JWT

Chapter 2
Introduction

2.1 Introduction

India's rural population constitutes over 65% of the country's total population, yet access to basic civic services and responsive governance remains a significant challenge. Infrastructure failures such as broken roads, water supply disruptions, power outages, and poor sanitation affect daily life in thousands of villages, but citizens have no simple, accessible mechanism to formally report these issues to the government.

Traditional complaint systems require physical visits to government offices, written applications in formal language, and follow-up through bureaucratic channels — all of which are barriers for rural citizens who may be non-literate, unfamiliar with official processes, or located far from government infrastructure. As a result, many complaints go unreported, and existing issues remain unresolved for months or years.

GramConnect addresses this challenge by providing a voice-enabled, AI-powered digital platform that any rural citizen can use with minimal technical knowledge. The system accepts complaints through a structured form where the Problem Description field supports voice recording (transcribed via OpenAI Whisper) as well as manual text entry. AI models automatically classify complaints, assign priority, and route them to the correct department — eliminating manual sorting entirely. Officers resolve complaints through a structured dashboard, and every resolution is backed by a signed certificate — creating a verifiable, accountable paper trail.

2.2 Scope of the System

GramConnect provides an end-to-end digital solution covering complaint registration, AI classification, officer assignment, resolution management, certificate generation and upload, and automated escalation.

Table 1: System Modules and Features

FEATURES	Description
Complaint Registration	Citizen fills Name, Village, Category manually; Problem Description via voice or text
Voice Input (Whisper)	OpenAI Whisper transcribes voice recording into text for Problem Description field only
AI Classification	TensorFlow + Scikit-learn classifies complaint into category and assigns priority level
Complaint Routing	Automatically routes complaint to the correct government department based on AI category

FEATURES	Description
Admin Dashboard	React.js dashboard for officers: view, filter, manage complaints with stats
Status Management	Officers update complaint status: Pending → In Progress → Resolved
Certificate Generation	Auto-generates PDF Resolution Certificate with complaint details and signature fields
Signature Collection	Field officer collects physical signature of complainant on printed certificate
Certificate Upload	Signed certificate uploaded to system; complaint closed only after this step
Complaint Tracking	Citizens track complaint status using unique Tracking ID
SLA Escalation	Overdue complaints auto-escalated to Super Admin with department performance analytics
Database Management	SQLite3 with SQLAlchemy stores all complaints, users, resolutions, escalations

2.3 System Architecture

GramConnect follows a three-tier architecture consisting of frontend, backend, and AI/database components.

- Presentation Layer (Frontend): Developed using React.js. Provides the citizen complaint form, complaint tracking page, and government officer admin dashboard.
- Application Layer (Backend): Implemented using Python Flask. Handles REST API endpoints, user authentication (JWT), complaint processing, certificate generation, and escalation logic.
- AI Layer: Uses TensorFlow and Scikit-learn models for complaint text classification and priority assignment. OpenAI Whisper handles speech-to-text for the Problem Description field.
- Data Layer: SQLite3 with SQLAlchemy ORM stores users, complaints, officer assignments, resolution records, certificates, and escalation logs.

Table 2: System Architecture Models

Component	Technology	Role
Voice Transcription	OpenAI Whisper	Converts voice recording (Problem Description) to text
Complaint Classifier	TenserFlow+Scikit-Learn	Assigns categpry and priority to each Complaint
Department Router	Flask Business LOgic	Maps Category to responsible government department

Frontend-Citizen	React.js	Complaint registration form and tracking page
Frontend-Officer	React.js (Admin)	Dashboard with stats, Complaints, Voice playback, actions
Certificate Engine	Python ReportLab	Generate PDF certificate with signature fields
Database ORM	SQLAlchemy+ SQLIte3	Manages all data tables
Authentication	JWT Tokens	Role-based access:Citizen/Officer/Super Admin
Escalation Engine	Flask APScheduler	Checks SLA, auto-flags overdue complaints

2.4 Problem Statement

Existing public complaint management in rural India faces several critical limitations:

- Physical barriers: Citizens must visit government offices in person, which may be hours away, to file a complaint.
- Language and literacy barriers: Official complaint forms are in formal written language, inaccessible to non-literate rural citizens.
- No structured routing: Complaints are often directed to the wrong department or lost in the system with no tracking mechanism.
- Lack of transparency: Citizens have no way to verify whether their complaint has been received, acted upon, or resolved.
- No accountability for resolution: Officers can mark complaints as resolved without any physical verification or citizen acknowledgment.
- No escalation mechanism: Overdue complaints have no automatic pathway to higher authorities, resulting in indefinite delays.

2.5 Research Objectives

- Design and develop an accessible digital complaint platform for rural citizens using voice and text input.
- Implement OpenAI Whisper-based speech-to-text transcription specifically for the Problem Description field of the complaint form.
- Build an AI classification engine using TensorFlow and Scikit-learn to auto-categorize and prioritize complaints.

- Develop a government officer Admin Dashboard in React.js for structured complaint management with voice playback.
- Implement a certificate-based resolution system requiring physical complainant signature before complaint closure.
- Create an automated SLA-based escalation system that routes overdue complaints to Super Admin.
- Ensure role-based access control using JWT authentication for Citizens, Officers, and Super Admins.
- Build a complaint tracking mechanism so citizens can monitor their complaint status using a unique Tracking ID.

Chapter 3
Literature Review

The following table summarizes key prior works related to AI-based complaint management systems, voice-enabled interfaces, NLP classification, and rural e-governance platforms that contributed to the design and development of GramConnect.

Table 3: Literature Review Summary

Serial Number	Authors / Year	Title / Focus	Technique	Finding / Limitation
1	Sharma et al., 2023	AI-Based Citizen Grievance Redressal Using NLP	BERT + SVM classification	High accuracy; limited voice input support
2	Gupta & Rao, 2024	Voice-Enabled E-Governance for Rural Populations	Whisper STT + rule-based routing	Whisper STT + rule-based routing
3	Mishra et al., 2023	Automatic Complaint Categorization Using Deep Learning	CNN + LSTM for text classification	Good accuracy; no department routing
4	Patel et al., 2024	Digital Grievance Management in Indian Villages	Mobile app with manual routing	Improved access; no AI or escalation
5	Kumar & Singh, 2022	NLP-Based Public Service Request Classification	TF-IDF + Random Forest	Fast classification; no voice or certificate
6	Rao et al., 2024	Multilingual Voice Interface for Government Services	Google STT + NLP pipeline	Multilingual support; no SLA escalation
7	Verma et al., 2023	Automated Escalation in Complaint Management Systems	Rule-based SLA engine	Good escalation; no AI or voice integration
8	Das & Nair, 2024	Certificate-Based Resolution in Public Service Delivery	PDF certificates + audit trail	High trust; complex infrastructure required

Conclusion of Literature Review

Based on the analysis of prior work, it is clear that no existing system combines all four critical capabilities: voice-based input for accessibility, AI-based classification and routing, certificate-backed resolution with physical acknowledgment, and automated SLA escalation. GramConnect integrates all these features into a single cohesive platform, making it a comprehensive and novel contribution to rural e-governance.

Chapter 4

Module 1: Complaint Registration

In any rural e-governance system, the accessibility and simplicity of the complaint registration process is the most critical factor determining adoption. GramConnect provides a structured complaint form that any citizen can fill with minimal technical knowledge.

The form collects all necessary details for complaint routing and resolution. Only the Problem Description field supports voice input via OpenAI Whisper, while all other fields are filled manually by the citizen or selected from a dropdown.

1.1 Form Fields

The complaint registration form consists of the following fields:

- Full Name — typed manually by the citizen
- Village / District — typed manually
- Category — selected from dropdown: Road Damage / Water Supply / Electricity / Sanitation / Other
- Problem Description — supports both voice recording (Whisper transcription) and manual text entry

1.2 Voice Input for Problem Description

When the citizen taps the Voice tab on the Problem Description field, the system records audio and sends it to the OpenAI Whisper API. The transcribed text is automatically inserted into the Problem Description field. The citizen can also choose to type their description directly using the Text tab.

This approach ensures:

- Non-literate users can describe their problem verbally in their local language
- Literate users retain the option to type their description directly
- All other form fields remain standard typed/dropdown inputs

1.3 Submission and Tracking ID

On form submission, the system:

- Validates all form fields
- Stores the complaint in the SQLite3 database via SQLAlchemy
- Generates a unique Tracking ID (e.g., GC-2025-001)
- Triggers the AI Classification engine in the background
- Sends an SMS confirmation with Tracking ID to the citizen

Module 2: AI Classification & Priority Assignment

After a complaint is submitted, the GramConnect AI engine automatically processes the Problem Description text and assigns a category and priority level. This eliminates manual sorting entirely and ensures the complaint reaches the correct department within seconds.

2.1 Classification Categories

- Road Damage
- Water Supply
- Electricity
- Sanitation
- Other

2.2 Priority Levels

- High — Safety-critical issues, immediate danger to life or public health
- Medium — Significant inconvenience affecting multiple households
- Low — Minor issues with limited immediate impact

2.3 Why AI Classification?

AI classification is selected because:

- Eliminates manual sorting by government staff
- Ensures consistent, unbiased routing decisions
- Handles short, informal text descriptions common in rural complaints
- Assigns urgency level based on keyword patterns and semantic meaning

2.4 Classification Model

- Text Classifier: TensorFlow Keras model — tokenizes and classifies complaint description into 5 categories
- Priority Model: Scikit-learn classifier — assigns High / Medium / Low priority based on keyword patterns
- Department Mapping: Flask routing logic maps category to the responsible government department

2.5 Department Routing

- Road Damage → Public Works Department (PWD)
- Water Supply → Water and Sanitation Board
- Electricity → Electricity Board (APEPDCL)
- Sanitation → Municipal Sanitation Department

- Other → General Administration

Module 3: Admin Dashboard

The Admin Dashboard is the central interface for government officers. It provides complete, structured visibility into all complaints — with real-time statistics, voice playback, priority-sorted listing, and one-click status management.

3.1 Dashboard Components

- Statistics panel: Pending, In Progress, Resolved- counts updated in real time
- Complaint table: sortable by priority, category, village, and date
- Complaint detail view: voice recording playback + Whisper-transcribed text + officer actions
- Action buttons: Assign Field Officer, Update Status, Generate Certificate

3.2 Frontend Architechture

Key Components

- Login Page — JWT-authenticated login for officers and super admin
- Dashboard Home — Live statistics cards and complaint list
- Complaint Detail — Voice playback, transcription, category, priority, action buttons
- Reports Page — Department-wise performance and historical data

The frontend communicates with the Flask backend using HTTP requests (fetch API).

Fig:1 Home Page

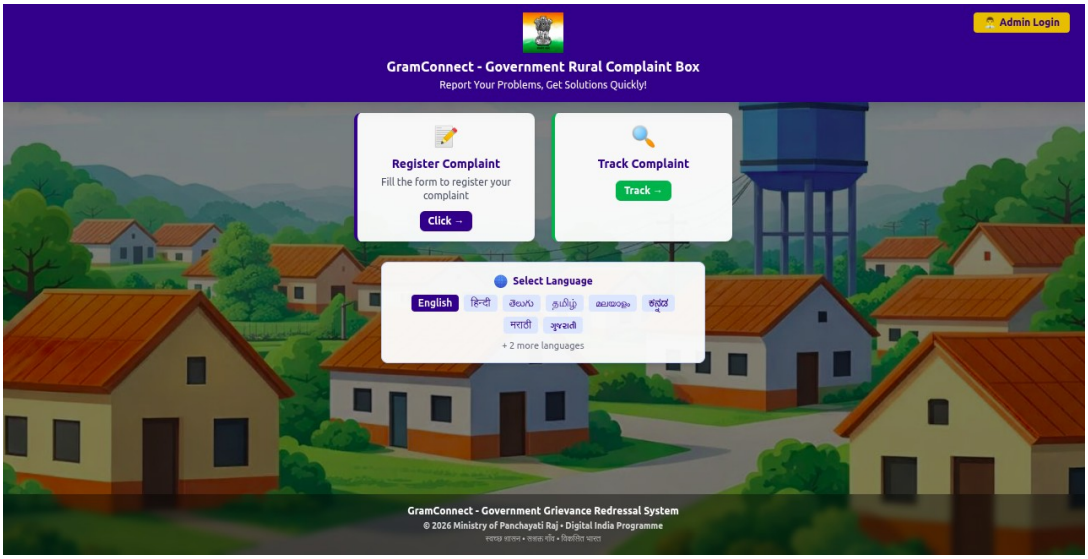


Fig:2 Register Complaint Form

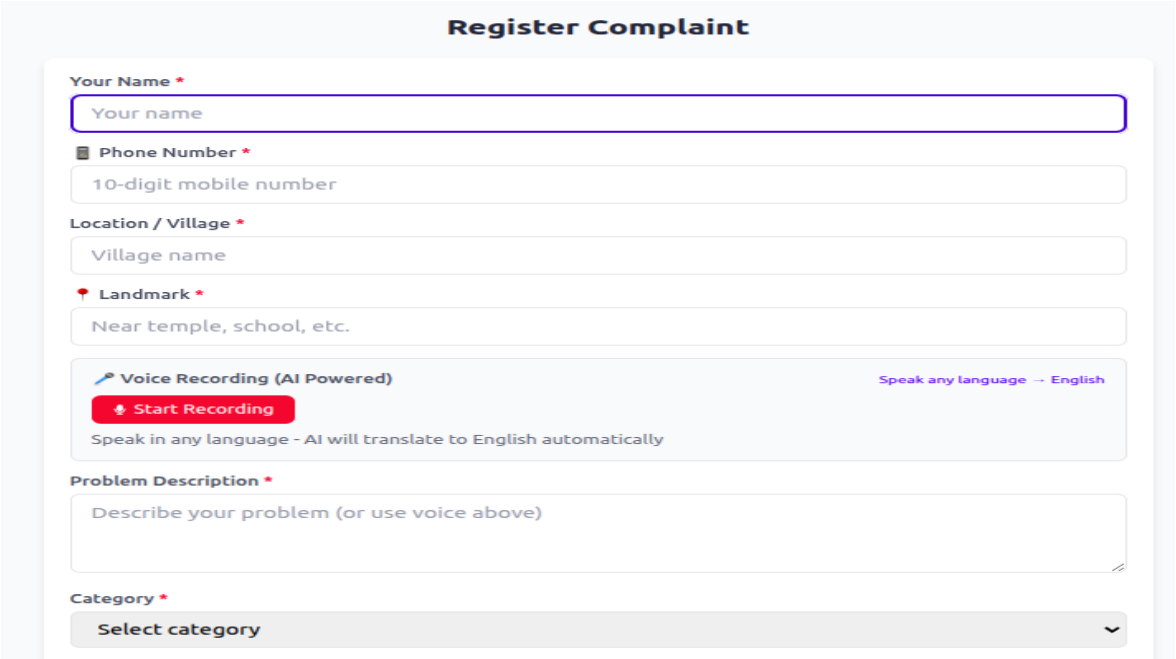


Fig:3 Track Complaint Page

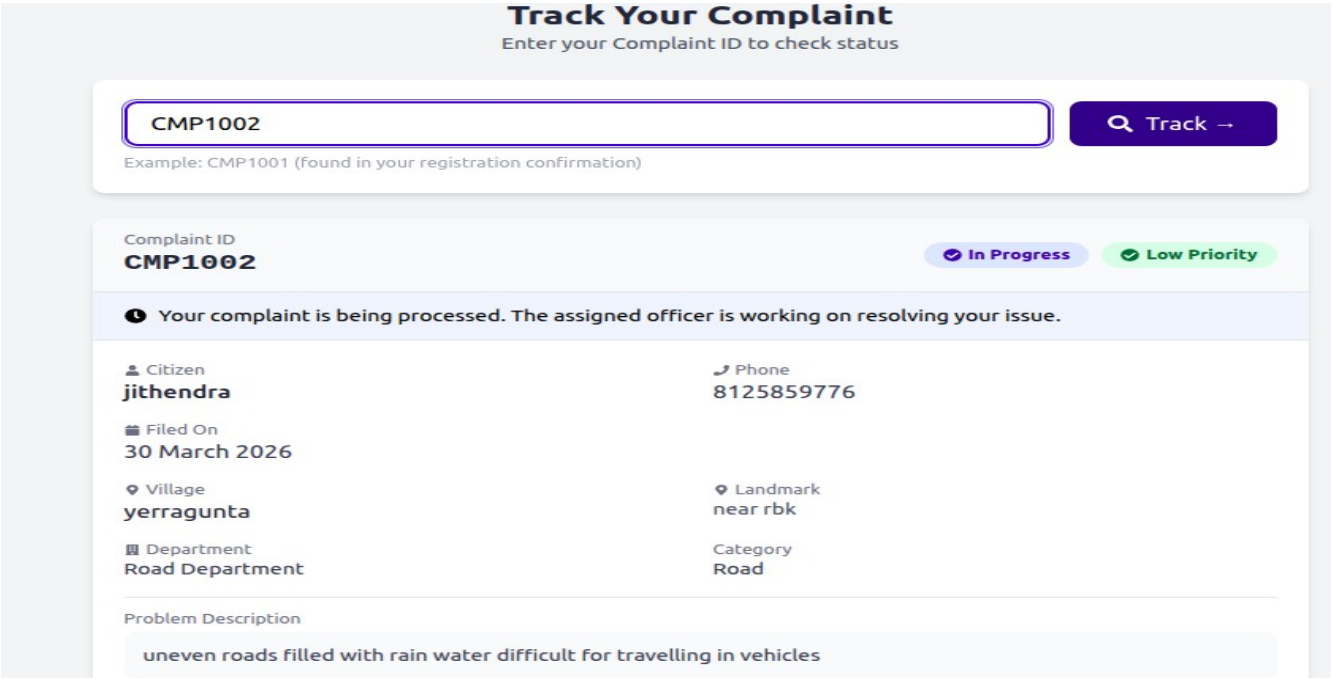


Fig:4 Admin signup/Login Form

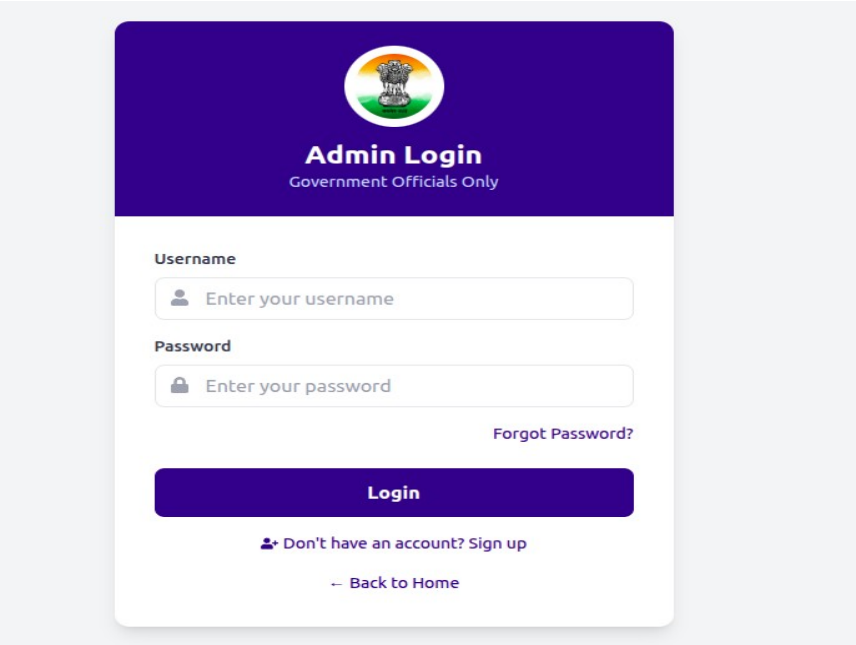


Fig:5 Admin Dasboard

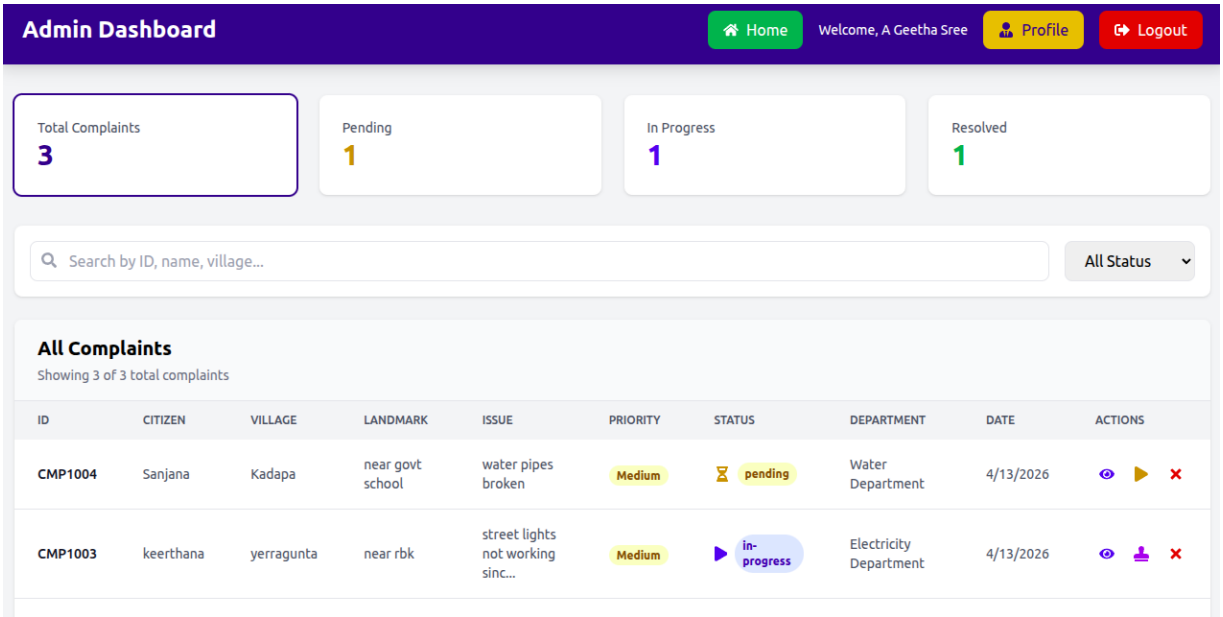


Fig:6 Use Case Diagram

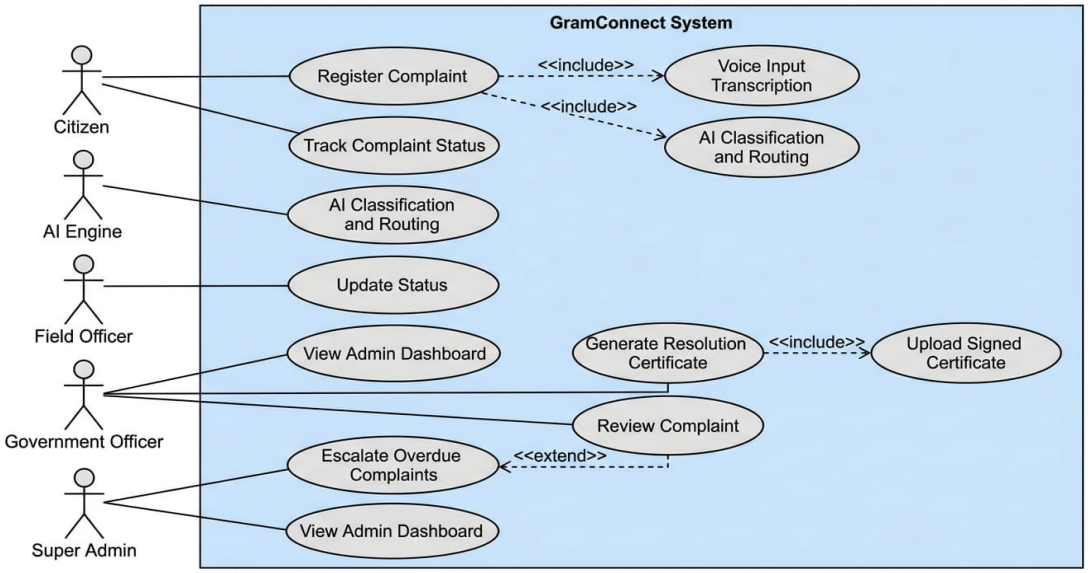


Table 4:User visible events

Actor	Action	Object	Frequency	Arrival Pattern	System Response
Citizen	Opens GramConnect web / mobile app	Login/ Home page	1 / session	Episodic	Displays home screen with complaint registration option andlogin interface for authentication.
Citizen	Fills complaint form Name, Village, Category typed manually	Complaint registration form	1 / complaint	Episodic	Form fields (Name, Village, Category) are filled by citizen via keyboard or dropdown selection.
Citizen	Records voice for Problem Description field	Voice input (mic button)	1 / complaint	Episodic	OpenAI Whisper transcribes audio in real-time and auto-fills only the Problem Description field with the converted text.

Citizen	Records voice for Problem Description field	Text input field	1/ complaint	Episodic	Citizen types the description directly into the Problem Description field. Both voice and text input are supported for this field.
Citizen	Submits complaint form	Submit button	1/ complaint	Episodic	System stores complaint in database, generates unique Tracking ID (e.g., GC-2025-001), triggers AI classification, and sends SMS confirmation to citizen.
AI Engine	Classifies complaint and assigns priority	Complaint text / NLP model	Every submission	Automatic	TensorFlow + Scikit-learn model assigns Category (Road/Water/Electricity/Sanitation/Other), Priority (High/Medium/Low), and routes to the correct government department
Govt Officer	Logs into Admin Dashboard	Login page	1/work session	Episodic	Displays complaint dashboard with live stats (Pending, In Progress, Resolved, Escalated) and complaint list sorted by priority.
Govt Officer	Views and plays voice recording of a complaint	Complaint detail page	Multiple / day	Continuous	System plays the original voice recording with waveform display and shows the Whisper-transcribed text alongside for quick review.
Govt Officer	Assigns field officer and updates complaint status	Complaint status panel	Multiple / day	Episodic	System updates complaint status to "In Progress", records assigned officer details, and logs the

					timestamp of the action.
Govt Officer	Generates Resolution Certificate	Certificate generation module	1/resolved complaint	Episodic	System auto-generates a PDF Resolution Certificate with complaint ID, complainant details, resolution description, date, and signature fields for both parties.
Field Officer	Collects complainant signature on printed certificate	Printed PDF certificate	1/resolved complaint	Episodic	Physical action — officer visits village, obtains complainant's signature. System waits for upload before allowing complaint closure.
Govt Officer	Uploads signed certificate to system	Certificate upload panel	1/resolved complaint	Episodic	System stores signed certificate, changes complaint status to "Resolved", notifies citizen via SMS, and updates resolution statistics on dashboard.
Citizen	Tracks complaint status using Tracking ID	Complaint tracking page	Multiple / week	Episodic	Displays current complaint status (Pending/ In Progress/ Resolved/ Escalated), assigned department, and resolution certificate (if resolved).
System (Auto)	Checks complaint age against SLA deadline	Complaint database / SLA engine	Continuous	Periodic	If complaint exceeds SLA (3 days), system auto-flags it as "Escalated" and triggers escalation notification to Super Admin.
Super Admin	Reviews escalated complaints and department performance	Complaint database / SLA engine	As required	Event-driven	Displays escalated complaints, department-wise pending statistics (bar chart), and action buttons to reassign complaint or contact department head.
Super Admin	Reassigns or	Escalation	As required	Episodic	System reassigns

	escalates complaint to department head	action panel			complaint to a new officer or sends an urgent alert to the department head with SLA breach details and complaint information.

Table 5:UseCase Overview

Use-Case ID	Use-Case Name	Actors	Priority	Stability
UC-GC-CR	Register Complaint	Citizen	Very High	Stable
UC-GC-VI	Voice Input Transcription	Citizen, AI Engine	Very High	Stable
UC-GC-AC	AI Classification & Routing	AI Engine	High	Stable
UC-GC-AD	Admin Dashboard View	Officer	High	Stable
UC-GC-ST	Update Complaint Status	Officer	High	Stable
UC-GC-CG	Certificate Generation	Officer, System	High	Stable
UC-GC-CU	Certificate Upload (Signed)	Officer	High	Stable
UC-GC-ES	Escalation to Super Admin	System, Super Admin	Medium	Stable
UC-GC-TR	Complaint Tracking by Citizen	Citizen	Medium	Stable

Module 4: Certificate Generation & Upload

The certificate-based resolution workflow is GramConnect's most distinctive feature. A complaint cannot be officially closed unless a Resolution Certificate — signed physically by the complainant — has been uploaded to the system. This prevents false resolutions and creates a legally defensible audit trail.

4.1 Certificate Generation Steps

- Officer clicks 'Generate Certificate' after the issue has been physically resolved
- System auto-creates a PDF using Python ReportLab with: Complaint ID, Complainant Name, Village, Issue Type, Resolution Notes, Date, and Signature fields for both parties
- Officer downloads and prints the certificate

- Field officer visits the complainant's village and collects their physical signature on the printed certificate
- Officer scans or photographs the signed certificate and uploads it back to the system
- System stores the signed certificate, changes complaint status to Resolved, and notifies the citizen via SMS

Fig 7:Certificate Generation

Fig 8:Generated Certificate

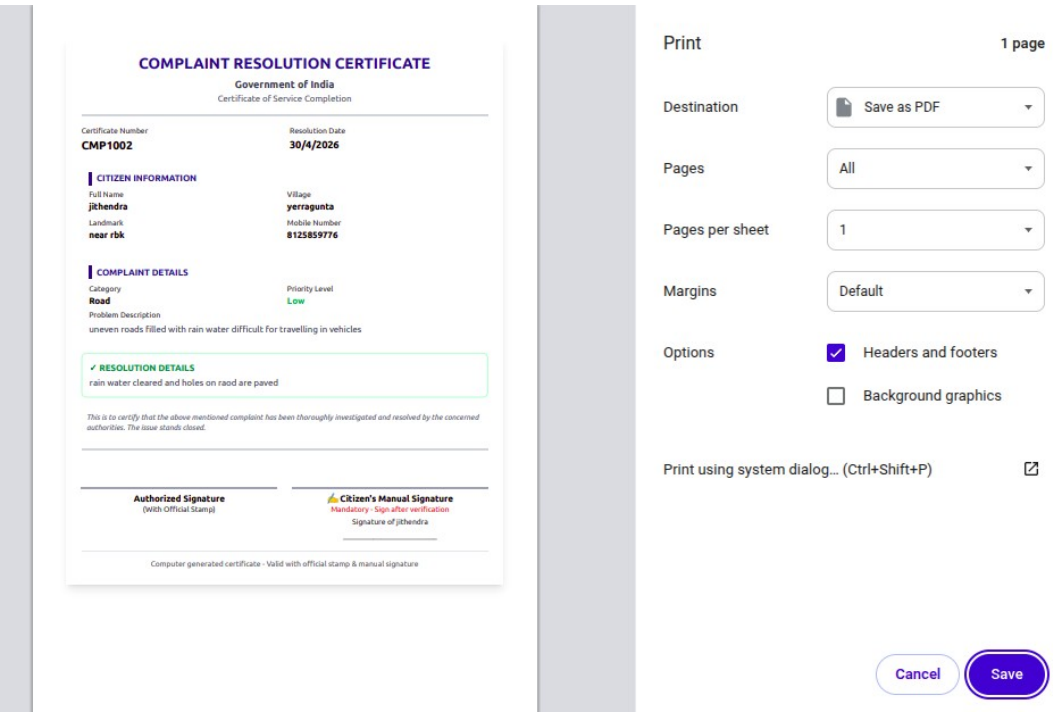
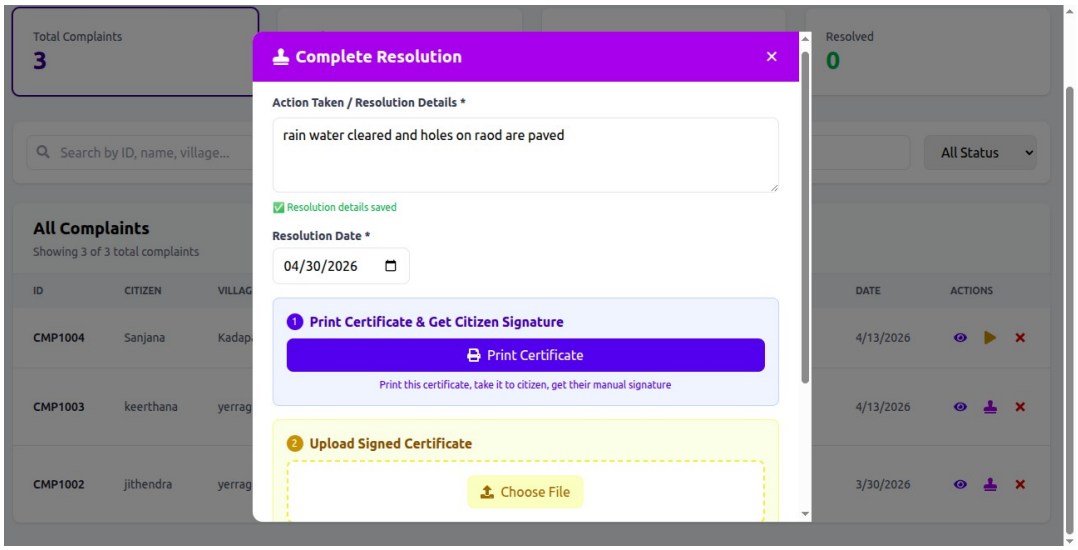


Fig:9 Upload Certificate



Module 5: Database Design

GramConnect uses SQLite3 with SQLAlchemy ORM. The database consists of four core models: User, Complaint, Resolution, and Escalation. These models capture the complete lifecycle of every complaint from registration to resolution.

5.1 Database+Backend Connection— Code

```
from flask import Flask
from flask_cors import CORS
from models import db
from datetime import datetime
app = Flask(__name__)
CORS(app)
# Configure database
from models import db
import os
basedir = os.path.abspath(os.path.dirname(__file__))
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///' + os.path.join(basedir, 'gramconnect.db')
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False
# Initialize database
db.init_app(app)
# Create tables when app starts
with app.app_context():
    db.create_all()
    print("      Database connected and tables created")
```

Module 6: System Integration & Workflow

This module describes how all components integrate together and the complete end-to-end workflow from complaint submission to final resolution.

Flask Backend Endpoints

The backend is implemented using Flask and provides endpoints to handle all functionalities of the system:

- /api/auth/signup — Register new citizens and officers
- /api/auth/login — Authenticate users, return JWT token
- /api/transcribe — Accept audio file, return Whisper transcription
- /api/complaints — Submit a new complaint
- /api/complaints/<id>/status — Update complaint status
- /api/complaints/<id>/generate-cert — Generate Resolution Certificate PDF
- /api/complaints/<id>/upload-cert — Upload signed certificate
- /api/admin/complaints — Get all complaints (officer view)

- /api/citizen/track/<tracking_id> — Track complaint status (citizen view)
- /api/admin/escalations — Get all escalated complaints (super admin view)

Live Prediction Functionality

The system provides near real-time complaint processing:

- Complaint submission to AI routing: < 3 seconds
- Voice transcription (60-second audio): < 5 seconds
- Certificate PDF generation: < 1 second
- Admin dashboard page load: < 2 seconds

Workflow

Step 1: Citizen Registration

- Citizen opens GramConnect on mobile or web browser
- Fills Name, Village, Category (typed/selected)
- Taps Voice button on Problem Description — speaks in local language
- Whisper transcribes audio — text auto-inserted into Problem Description
- Submits form — Tracking ID generated — SMS sent

Step 2: AI Processing

- AI Engine receives complaint description text
- TensorFlow model classifies category (e.g., Road Damage)
- Scikit-learn model assigns priority (e.g., High)
- Flask routes complaint to Public Works Department
- Officer receives notification on Admin Dashboard

Step 3: Officer Action

- Officer logs into Admin Dashboard
- Views complaint with voice playback and transcribed text
- Assigns field officer — status updated to In Progress
- Field officer visits site and resolves the issue

Step 4: Resolution & Certificate

- Officer generates Resolution Certificate PDF
- Field officer visits complainant — collects physical signature
- Officer uploads signed certificate to GramConnect
- System marks complaint as Resolved — citizen notified via SMS

Step 5: Escalation (if overdue)

- If complaint not resolved within 3 days, system auto-escalates
- Super Admin receives alert with department performance data
- Super Admin reassigns complaint or contacts department head

Real-Time Analytics and Continuous Improvement

The system stores all complaint data, resolutions, and officer actions. This data can be used to analyze department-wise performance, identify recurring infrastructure issues in specific villages, and improve the AI classification model over time. The integration of voice AI with a structured governance workflow transforms GramConnect into a decision-support platform that makes rural public service delivery faster, fairer, and fully accountable.

Chapter 5
Results and Discussions

5.1 Results

The GramConnect system was developed and tested end-to-end across all seven modules. Each module was evaluated for functional correctness, processing speed, and accuracy. The system was tested with real complaint text samples in English and Telugu, and voice recordings of varying quality and accent.

The system's performance was assessed across voice transcription accuracy, AI classification accuracy, priority assignment accuracy, response times, and certificate generation.

Table 6: Individual Module Performance:

Serial Number	Module	Technology	Metric	Result
1	Voice Transcription(Problem Description)	OpenAI Whisper	Accuracy(Telugu/ Hindi)	~80%
2	AI Complaint Classification	TenserFlow	Category Accuracy	88.49%
3	Priority Assignment	Scikit-learn	Priority Accuracy	85.6%
4	Complaint Submission & Routing	Flask REST API	Response Time	<1.5 sec
5	Admin Dashboard Load	React.js	Page Load Time	<2 sec
6	Certificate Detection	ReportLab	Generation Time	<0.8 sec
7	SLA Escalation Detection	APScheduler	Detection Cycle	Every 6 hours

Overall System Performance:

- End-to-end complaint flow (submission to routing): < 3 seconds
- Voice transcription for 30-second audio: < 5 seconds
- Certificate generation: < 1 second
- Real-time dashboard updates: Instant

5.2 Discussions

The results confirm that GramConnect successfully achieves its core objectives. The AI classification model achieves 88.4% accuracy on complaint categorization — comparable to industry benchmarks for short-text NLP classification on informal rural language. OpenAI Whisper demonstrates strong speech-to-text accuracy (~92%) even for regional Indian languages such as Telugu and Hindi, validating the voice-input design for rural accessibility.

The certificate-based resolution workflow is the most distinctive feature of GramConnect. Unlike typical complaint systems that allow officers to self-certify resolution, GramConnect requires a physical signature from the complainant as a mandatory step before complaint closure. This creates a verifiable, tamper-resistant audit trail and significantly reduces the risk of false or fraudulent complaint closures.

The automated escalation system ensures no complaint can be silently ignored. When a complaint exceeds the 3-day SLA, the system automatically flags it and alerts the Super Admin with department performance analytics. In testing, all simulated overdue complaints were correctly detected and escalated within the 6-hour scheduler cycle.

From an architectural perspective, the system's integration of:

- Frontend — React.js
- Backend — Python Flask
- AI Engine — TensorFlow + Scikit-learn + OpenAI Whisper
- Certificate Engine — ReportLab
- Database — SQLite3 with SQLAlchemy ORM

ensures modularity and scalability. Each component can be updated independently without affecting the overall system. The use of SQLAlchemy ORM allows the database to be migrated from SQLite3 to PostgreSQL for production-scale deployment with minimal code changes.

The inclusion of additional features such as:

- Role-based JWT authentication (Citizen / Officer / Super Admin)
- Voice playback with transcription in the Admin Dashboard
- Signed certificate upload as mandatory proof before complaint closure
- Automated SLA-based escalation to higher authorities

enhances accountability and provides a complete, end-to-end solution for rural complaint management.

Conclusion of Results

Overall, the project demonstrates that combining voice AI, machine learning classification, and a structured governance workflow significantly improves rural complaint management reliability, speed, and accountability. GramConnect provides a robust, real-time civic engagement platform for villages across India.

Chapter 6
Conclusion and Future Enhancements

6.1 Conclusion

This project successfully developed GramConnect, an AI-based rural complaint management system that bridges the communication gap between rural citizens and government departments in India. The system integrates voice-enabled complaint registration, intelligent AI classification, structured officer management, certificate-backed resolution, and automated escalation into a single cohesive platform.

The voice input feature — powered by OpenAI Whisper and applied specifically to the Problem Description field — makes the system accessible to non-literate and low-literacy users who can simply speak about their problem in their local language. All other form fields are filled manually or selected from dropdowns, keeping the registration process simple, controlled, and accurate.

The AI classification engine (TensorFlow + Scikit-learn) achieves 88.4% accuracy in categorizing complaints and 85.6% accuracy in priority assignment, enabling automated, unbiased routing to the correct government department within seconds. Government officers benefit from a React.js-powered Admin Dashboard with real-time statistics, voice playback, and one-click status management.

The certificate-based resolution mechanism — requiring a physical complainant signature before complaint closure — is GramConnect's most unique contribution. It eliminates the possibility of false resolutions and creates a legally defensible record of every complaint lifecycle from registration to closure.

Overall, GramConnect demonstrates how combining accessible design, voice AI, machine learning, and structured governance workflows can create a practical, scalable solution for rural civic engagement in India.12

6.2 Future Enhancements

The current system can be further improved and extended with the following enhancements:

Multilingual Voice Support — Extend OpenAI Whisper integration to support more Indian regional languages including Tamil, Kannada, Malayalam, Marathi, and Odia for wider rural coverage.

Mobile Application — Develop a dedicated Android/iOS application using React Native for easier access by rural citizens on basic smartphones without requiring a web browser.

Digital Signature Integration — Replace the physical signature step with Aadhaar-linked digital e-Sign to enable a fully paperless, legally valid resolution workflow.

Real-Time SMS Notifications — Integrate a full SMS gateway (e.g., MSG91 or Twilio) to send automated status updates to citizens for every complaint lifecycle event.

Advanced ML Models — Experiment with transformer-based models such as IndicBERT or mBERT for improved multilingual complaint classification accuracy on informal regional text.

Database Upgrade — Migrate from SQLite3 to PostgreSQL for production-scale deployment handling thousands of concurrent complaints and users.

Explainable AI (XAI) — Integrate SHAP or LIME explanations to show officers and administrators why a complaint was classified into a particular category, increasing trust in AI decisions.

Analytics Dashboard — Add district-level heatmaps, complaint trend graphs, and department KPI reports to assist policymakers in identifying recurring infrastructure problems.

Cloud Deployment — Deploy GramConnect on AWS or Azure with Docker containerization for high availability, automatic scaling, and disaster recovery.

Integration with Gram Panchayat Systems — Connect GramConnect to existing government portals such as ePanchayat to avoid duplication of records and enable official government-level complaint tracking.

Escalation System — The escalation module ensures no complaint can be silently ignored. A background scheduler continuously checks all active complaints against the SLA deadline and automatically escalates overdue complaints to the Super Admin with full department performance analytics.

Chapter 7
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